

Immunodeficiency 118 (IMD118): A Comprehensive Disease Characteristics Report

Disease: Immunodeficiency 118 (IMD118) **MONDO ID:** MONDO:0958030 · **OMIM:** #301115 · **Gene:** *MCTS1* (Xq24) **Category:** Mendelian — X-linked recessive inborn error of immunity (a subtype of Mendelian Susceptibility to Mycobacterial Disease, MSMD) **Primary reference:** Bohlen J, Zhou Q, Philippot Q, et al. "Human MCTS1-dependent translation of JAK2 is essential for IFN- γ immunity to mycobacteria." 2023 ([PMID: 37875108](#)).

Summary

Immunodeficiency-118 (IMD118) is an ultra-rare, X-linked recessive inborn error of immunity caused by hemizygous loss-of-function mutations in *MCTS1* on chromosome Xq24. The disease was defined by a single landmark study — Bohlen, Zhou, Béziat, Casanova and colleagues (2023, [PMID: 37875108](#)) — which identified complete MCTS1 deficiency in 5 **unrelated males** from kindreds of different ancestries (China, Finland, Iran, and Saudi Arabia). Clinically, IMD118 belongs to the MSMD spectrum: affected males present in infancy with disseminated infection by weakly virulent mycobacteria, most commonly after *Bacillus Calmette-Guérin* (BCG) vaccination, while otherwise displaying essentially normal immunity, growth, and development.

The mechanism is a specific and elegant translational defect. MCTS1 (Malignant T-cell-amplified sequence 1) is a translation re-initiation and 40S ribosome-recycling factor that forms a heterodimer with DENR. Complete MCTS1 deficiency impairs the translation of a small subset of proteins — most importantly the tyrosine kinase **JAK2** — in all cell types tested. JAK2 loss cripples signaling downstream of **IL-23** (and partially IL-12), which in turn reduces **IFN- γ** production by innate-like adaptive **MAIT (mucosal-associated invariant T)** and **$\gamma\delta$ T lymphocytes** upon mycobacterial challenge. Because IFN- γ is the central macrophage-activating cytokine controlling intracellular mycobacteria, its failure produces the

characteristic mycobacterial susceptibility. The selectivity of the JAK2 defect is explained by two ultra-short "start-stop" upstream open reading frames (**stuORFs**) in the *JAK2* 5'UTR that render JAK2 translation uniquely dependent on MCTS1-mediated ribosome recycling.

Prognosis is generally favorable but not benign. Among the 5 reported patients, one died of disseminated mycobacterial disease while the other four responded to antimycobacterial therapy and remained asymptomatic after early childhood, with normal growth and development. Critically, patient leukocytes retained **normal responses to exogenous IFN- γ and IFN- α** , meaning the block is upstream of the IFN- γ receptor and is potentially bypassable — providing a clear rationale for recombinant IFN- γ as adjunctive therapy, as used successfully in other MSMD genotypes. A notable paradox is that *MCTS1* is an established oncogene, yet its complete germline loss produces only a narrow immunological phenotype rather than developmental defects or cancer, indicating that its translational function is largely physiologically redundant except along the IL-23→JAK2→IFN- γ axis.

Key Findings

Finding 1 — IMD118 is X-linked recessive MCTS1 deficiency causing isolated mycobacterial disease

IMD118 (OMIM #301115) is caused by **hemizygous loss-of-function variants in *MCTS1*** (Xq24) and was described in 5 unrelated males from four countries. As the primary report states: "*We report X-linked recessive MCTS1 deficiency in men with mycobacterial disease from kindreds of different ancestries (from China, Finland, Iran, and Saudi Arabia)*" (PMID: 37875108).

The five patients carried a spectrum of loss-of-function variant types: **2 frameshift variants, 1 splice-site variant, and 2 in-frame 3' deletions**, the latter two both yielding a truncated protein lacking residues Ala133–Lys181 (p.Ala133_Lys181del). All variants result in complete MCTS1 deficiency. The molecular consequence is defined by the authors: "*Complete deficiency of this translation re-initiation factor impairs the translation of a subset of proteins, including the kinase JAK2 in all cell types tested, including T lymphocytes and phagocytes.*" The downstream immunological consequence connects the translation defect to disease: "*Defective responses to IL-23 preferentially impair the production of IFN- γ by innate-like adaptive mucosal-associated invariant T cells (MAIT) and $\gamma\delta$ T lymphocytes upon mycobacterial challenge.*"

This establishes IMD118 as a single-gene, X-linked recessive disorder in which a general translation factor defect is funneled into a strikingly narrow clinical outcome via a single critical substrate (JAK2) and a single critical pathway (IL-23→IFN-γ).

Finding 2 — MCTS1 forms a heterodimer with DENR to mediate 40S ribosome recycling and translation reinitiation

MCTS1 (also called MCT-1; NCBI Gene 28985; Xq24) is a translation factor that partners with **DENR** (density-regulated reinitiation factor). Together MCTS1/DENR are the mammalian counterparts of the yeast Tma20/Tma22 heterodimer, with **eIF2D** (yeast Tma64) providing a parallel activity. These factors recycle post-termination 40S ribosomal subunits at stop codons and modulate reinitiation on mRNAs containing short upstream open reading frames (uORFs).

The functional role was demonstrated in yeast and in vitro systems: *"Tma64 (eIF2D), Tma20 (MCT-1), and Tma22 (DENR) function as 40S recycling factors in vitro"* (PMID: 30146315). The reinitiation-promoting property that is mechanistically central to IMD118 was shown by Jendruchová et al.: *"MCTS1/DENR enhance reinitiation at short upstream open reading frames (uORFs) harboring penultimate codons that confer dependence on these factors in bulk 40S recycling"* (PMID: 38903097). When MCTS1 is absent, 80S ribosomes queue behind stop codons and aberrant reinitiation occurs in 3'UTRs — the general failure that, at the *JAK2* locus, specifically prevents translation of the main *JAK2* open reading frame.

Finding 3 — MCTS1 is a known oncoprotein, yet germline complete loss causes only isolated mycobacterial disease

A striking feature of IMD118 is the mismatch between MCTS1's known oncogenic biology and the narrow phenotype of its germline loss. MCTS1/DENR are **established oncogenes**: *"DENR and MCTS1 have been identified as oncogenes in several different tumor entities. The heterodimeric DENR·MCTS1 protein complex promotes translation of mRNAs containing upstream Open Reading Frames (uORFs)"* (PMID: 35115540). The complex is cell-cycle regulated (DENR Ser73 phosphorylated by Cyclin B/CDK1 and Cyclin A/CDK2), and MCTS1 overexpression drives proliferation in laryngeal squamous cell carcinoma via the OTUD6B-LIN28B axis and LARP7 stabilization (PMID: 37634410, PMID: 35274760), and appears in breast-cancer prognostic gene signatures (PMID: 35441810, PMID: 35120331).

Despite this oncogenic role, the 5 hemizygous MCTS1-null males exhibited **only isolated mycobacterial disease** with otherwise normal immunity, growth, development, and no reported malignancy. The authors note: *"the lack of MCTS1-dependent translation re-initiation*

and ribosome recycling seems to be otherwise physiologically redundant in these patients" (PMID: 37875108). This tells us that, in humans, MCTS1's translational function is largely dispensable for viability and development, and that its physiological non-redundancy is confined to the anti-mycobacterial IFN- γ axis.

Finding 4 — IMD118 lies on the IL-23-to-IFN- γ axis, independently validated by IL23R and IL-23 deficiencies

IMD118 is mechanistically embedded within the well-characterized IL-12/IL-23/IFN- γ circuit whose disruption defines MSMD. The essential role of IL-23 was established by Philippot et al.: *"Human IL-23 is essential for IFN- γ -dependent immunity to mycobacteria"* (PMID: 36763636). Independent genetic validation comes from IL23R deficiency: a homozygous R381X mutation in *IL23R* causes MSMD, where *"impaired IL-23 immunity caused by a homozygous R381X mutation in IL23R underlies MSMD"* (PMID: 35829840), producing impaired IL-23-mediated STAT3 phosphorylation, reduced IFN- γ secretion, and isolated disseminated BCG/NTM disease.

The effector cells are innate-like T cells. MAIT cells recognize mycobacterial vitamin-B metabolites presented on MR1 and produce IFN- γ , TNF, and granzyme B, and are reduced/exhausted in active tuberculosis (PMID: 37147816, PMID: 41707313). MCTS1 deficiency impairs JAK2 $\rightarrow$ IL-23 signaling $\rightarrow$ IFN- γ from MAIT/ $\gamma\delta$ cells, placing IMD118 firmly in the same functional pathway as IL12B, IL12RB1, IL23R, IFNGR1/2, STAT1, and the other MSMD genes.

Finding 5 — Clinical spectrum: infantile-onset disseminated mycobacteriosis with recovery after early childhood

The OMIM #301115 Clinical Synopsis, derived from Bohlen et al. 2023, describes the clinical picture of the 5 unrelated males (aged 3–18 years at report). **Disseminated mycobacterial disease developed between 3 and 12 months of age, usually after BCG vaccination.** Reported organ-system features and suggested HPO terms include:

Feature	HPO term
Hepatomegaly	HP:0002240
Splenomegaly	HP:0001744
Osteomyelitis	HP:0002754
Fever	HP:0001945
Lymphadenopathy / enlarged lymph nodes	HP:0002716
Abscess	HP:0025615
Disseminated mycobacteriosis	(BCG/NTM)

Immunologically, patients had **normal numbers of circulating leukocyte subsets** — routine immune workup is typically unrevealing, a hallmark of MSMD that necessitates molecular diagnosis. Most patients recover with treatment and remain asymptomatic after early childhood. The source data underlie the OMIM synopsis: *"We report X-linked recessive MCTS1 deficiency in men with mycobacterial disease from kindreds of different ancestries"* (PMID: [37875108](#)).

Finding 6 — JAK2's selective MCTS1-dependence is caused by two ultra-short start-stop uORFs (stuORFs); exogenous IFN- γ responses remain intact

The molecular reason JAK2 is uniquely vulnerable among the many proteins translated in the cell was resolved by 5'UTR reporter assays. Among **17 MSMD/IFN- γ -immunity genes** tested in MCTS1-KO HeLa cells, **only JAK2 showed a >65% decrease in translation**, which was rescued by wild-type MCTS1 but not by patient variants or a synthetic loss-of-function control (A109D). The JAK2 5'UTR contains three uORFs, two of which are **ultra-short start-stop "stuORFs"** (uORF1, uORF2). In the absence of MCTS1, 40S ribosomes stall (a "roadblock") at these stuORF stop codons, blocking re-initiation at the JAK2 main ORF — confirmed directly by accumulation of 40S ribosomal footprints. CRISPR MCTS1-knockout in THP-1 monocytes reproducibly lowered endogenous JAK2 across 5 independent clones.

Functionally, patient leukocytes had **impaired IFN- γ production after BCG stimulation** and impaired IL-23 responses (but normal IL-12 responses), yet retained **NORMAL responses to exogenous IFN- γ and IFN- α** . This localizes the defect upstream of the IFN- γ receptor and predicts that exogenous IFN- γ can bypass the block.

The protein architecture underpinning MCTS1's role was characterized earlier: *"MCT-1 contains the PUA domain, a recently described RNA-binding domain that is found in several tRNA and rRNA modification enzymes... MCT-1 protein interacts with the cap complex through its PUA domain and recruits the density-regulated protein (DENR/DRP), containing the SUI1 translation initiation domain"* (PMID: 16982740).

Finding 7 — Prognosis is generally favorable but the disease can be fatal (1 of 5 patients died)

Of the 5 hemizygous MCTS1-deficient males, **one died of disseminated mycobacterial disease**; the remaining four responded to antimycobacterial treatment, had no other infections, and showed normal growth and development. The redundancy of MCTS1 outside the mycobacterial axis is emphasized by the authors: *"Surprisingly, the lack of MCTS1-dependent translation re-initiation and ribosome recycling seems to be otherwise physiologically redundant in these patients"* (PMID: 37875108). The preserved response to exogenous IFN- γ and IFN- α provides both a favorable prognostic indicator and a therapeutic target.

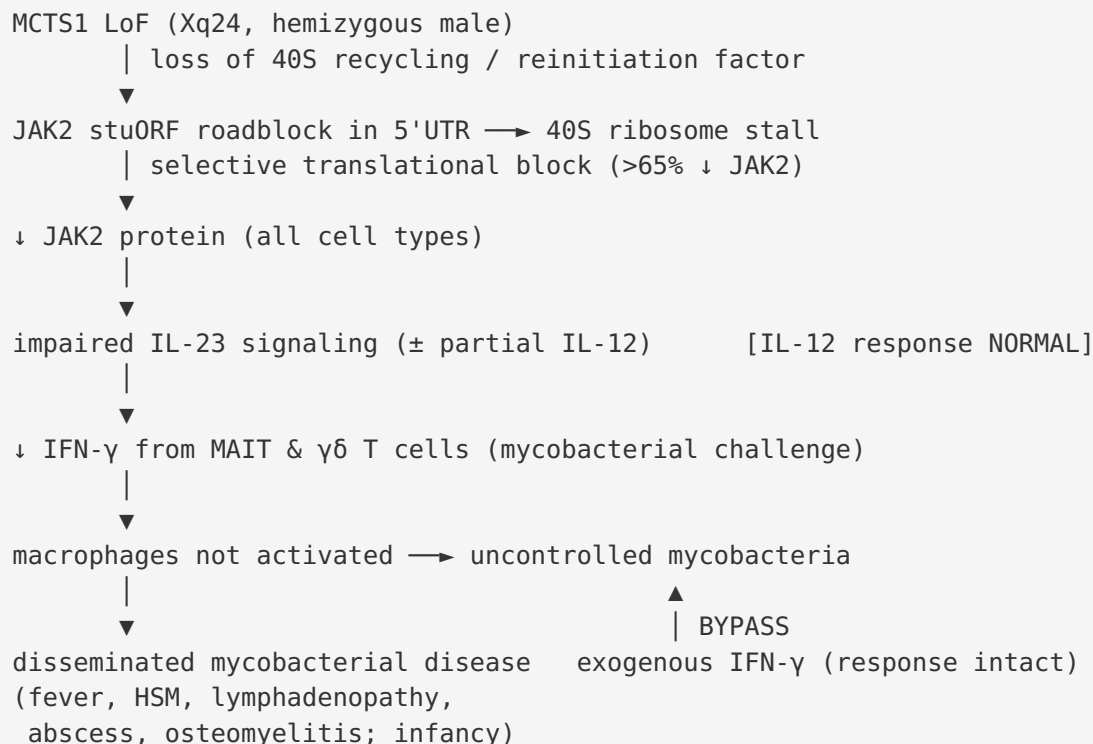
Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **hemizygous loss-of-function mutation in MCTS1** (Xq24; frameshift, splice-site, or in-frame 3' deletion) **leads to** complete absence of functional MCTS1 protein in a male (X-linked recessive; single X chromosome). *[demonstrated]*
2. Loss of MCTS1 **results in** failure of MCTS1/DENR-mediated 40S ribosome recycling and translation re-initiation. *[demonstrated in vitro and in cells]*
3. At the *JAK2* mRNA, the two ultra-short start-stop uORFs (stuORFs) in the 5'UTR **cause** 40S ribosomes to stall at the stuORF stop codons (a translational roadblock), which **blocks** re-initiation at the *JAK2* main ORF. *[demonstrated by 40S footprint accumulation and 5'UTR reporter assays]*
4. Blocked re-initiation **leads to** selectively reduced *JAK2* protein in all cell types, including T lymphocytes and phagocytes, while most other proteins are unaffected. *[demonstrated]*

5. Reduced JAK2 **results in** impaired signal transduction downstream of the IL-23 receptor (and partial impairment of IL-12 signaling). *[demonstrated — impaired IL-23 response, normal IL-12 response in patient cells]*
6. Impaired IL-23 signaling **preferentially impairs** IFN- γ production by innate-like adaptive **MAIT and $\gamma\delta$ T lymphocytes** upon mycobacterial challenge. *[demonstrated]*
7. Deficient IFN- γ **fails to** activate macrophages to control intracellular mycobacteria. *[inferred from the established MSMD paradigm — IFN- γ is the central macrophage-activating cytokine]*
8. Failure of macrophage activation **leads to** uncontrolled proliferation of weakly virulent mycobacteria (BCG, NTM), **resulting in** disseminated mycobacterial disease with fever, lymphadenopathy, hepatosplenomegaly, abscesses, and osteomyelitis in infancy. *[demonstrated clinically]*

Branch point (bypass): Because the lesion is upstream of the IFN- γ receptor, exogenous IFN- γ can activate macrophages directly (patient cells respond normally to IFN- γ and IFN- α) → therapeutic bypass and generally favorable prognosis.



Upstream vs downstream

- **Upstream / initiating:** MCTS1 loss → global translation-recycling defect (a general molecular lesion).
- **Bottleneck / specificity-determining:** JAK2 stuORFs — the single feature that channels a general translation defect into a narrow clinical phenotype.
- **Downstream / effector:** IL-23 signaling → MAIT/ $\gamma\delta$ IFN- γ → macrophage activation → mycobacterial control.

Ontology term suggestions

- **Gene/protein:** *MCTS1* (HGNC:7838), *JAK2* (HGNC:6192), *DENR*, *IL23R*, *IL12B*
- **GO biological process:** regulation of translational initiation (GO:0006446), positive regulation of interferon-gamma production (GO:0032729), response to molecule of bacterial origin, ribosomal subunit recycling
- **GO cellular component:** cytosolic ribosome (GO:0022626), cytosol (GO:0005829), messenger ribonucleoprotein / cap complex
- **CL cell types:** mucosal invariant T cell / MAIT (CL:0000940), gamma-delta T cell (CL:0000798), macrophage (CL:0000235), monocyte (CL:0000576)
- **UBERON anatomy:** liver (UBERON:0002107), spleen (UBERON:0002106), lymph node (UBERON:0000029), bone (UBERON:0001474)
- **NCIT treatment:** recombinant interferon gamma; antimycobacterial therapy; hematopoietic stem cell transplantation
- **Disease:** MONDO:0958030 (IMD118)

Section-by-Section Report Content

1. Disease Information

IMD118 is an X-linked recessive inborn error of immunity within the MSMD spectrum, characterized by selective susceptibility to weakly virulent mycobacteria (BCG vaccine strain and non-tuberculous mycobacteria). **Identifiers:** OMIM #301115; MONDO:0958030; gene *MCTS1* (Xq24). ICD-11 maps to inborn errors of immunity / MSMD categories; no dedicated

ICD-10 code. Synonyms: **MCTS1 deficiency**, **X-linked MCTS1 deficiency**, MSMD due to MCTS1 deficiency. The information is derived from an **aggregated disease-level primary report** (Bohlen et al. 2023, 5 individual patients) and OMIM, not from EHR-scale data.

2. Etiology

Causal factor: monogenic — hemizygous loss-of-function variants in *MCTS1*. **Genetic risk factor:** male sex (X-linked recessive; a single mutant X allele is sufficient in males). **Environmental trigger:** BCG vaccination and exposure to environmental (non-tuberculous) mycobacteria precipitate disease; the mutation is necessary, mycobacterial exposure is the environmental precipitant. No protective genetic variants are described; **exogenous IFN- γ acts as an acquired protective/therapeutic factor** because responses to it are intact. **Gene–environment interaction:** the classic MSMD interaction — an inherited IL-23→IFN- γ axis defect that becomes clinically manifest only upon mycobacterial exposure (especially live BCG vaccine).

3. Phenotypes

Predominantly **infectious/physical manifestations** rather than behavioral. Onset is **infantile (3–12 months)**, typically post-BCG. Features (with qualitative frequencies given the 5-patient series): disseminated mycobacteriosis (defining), fever (HP:0001945), lymphadenopathy (HP:0002716), hepatomegaly (HP:0002240), splenomegaly (HP:0001744), abscess (HP:0025615), osteomyelitis (HP:0002754). Severity is variable — from recoverable disseminated disease to fatal dissemination. Progression is **episodic/acute during active infection**, with recovery and asymptomatic status after early childhood in survivors. Quality-of-life impact is concentrated in infancy/early childhood during active infection; survivors have generally normal function thereafter.

4. Genetic / Molecular Information

Causal gene: *MCTS1* (Xq24; NCBI Gene 28985; HGNC:7838). **Variant types among 5 patients:** 2 frameshift, 1 splice-site, 2 in-frame 3' deletions (both → p.Ala133_Lys181del). **Classification:** pathogenic loss-of-function (ACMG). **Zygosity/origin:** germline, hemizygous in males. **Functional consequence:** complete loss of function → loss of 40S recycling/reinitiation → selectively reduced JAK2 translation. **Allele frequency:** extremely rare/private; not established as recurrent in gnomAD. **Modifier genes:** none formally identified. **Epigenetics/chromosomal abnormalities:** none reported; the mechanism is translational, not epigenetic.

5. Environmental Information

The key environmental factors are **mycobacterial exposures** — live attenuated **BCG vaccine** (*Mycobacterium bovis* BCG) and environmental **non-tuberculous mycobacteria (NTM)**. No toxin, radiation, pollution, occupational, or lifestyle factors are implicated. Infectious agents are causative triggers acting on the genetic substrate rather than independent causes.

6. Mechanism / Pathophysiology

See the **Mechanistic Model** section above for the full ordered causal chain. In brief: MCTS1 LoF → failure of 40S ribosome recycling/reinitiation → JAK2 stuORF roadblock → selective JAK2 depletion → impaired IL-23 (and partial IL-12) signaling → reduced IFN- γ from MAIT/ $\gamma\delta$ T cells → failed macrophage activation → disseminated mycobacterial disease. Molecular pathway: **JAK-STAT (JAK2) / IL-23→IFN- γ axis**. Cellular process: cytokine signaling and antimicrobial macrophage activation. Protein dysfunction: loss of function of a ribosome-recycling factor (MCTS1) causing loss of a downstream kinase (JAK2). Immune involvement: immunodeficiency (not autoimmunity).

7. Anatomical Structures Affected

Primary organs: reticuloendothelial/lymphoid system — **liver** (UBERON:0002107), **spleen** (UBERON:0002106), **lymph nodes** (UBERON:0000029), **bone/bone marrow** (osteomyelitis; UBERON:0001474). **Body systems:** immune/hematopoietic and, via sites of mycobacterial dissemination, potentially multi-organ. **Cell populations:** MAIT cells (CL:0000940), $\gamma\delta$ T cells (CL:0000798), macrophages/monocytes (CL:0000235/CL:0000576) — because the JAK2 defect is present in all cell types tested, including phagocytes. **Subcellular:** cytosolic ribosome/translation machinery (GO:0022626). **Lateralization:** not applicable (systemic/bilateral dissemination).

8. Temporal Development

Onset: infantile, 3–12 months, typically after BCG vaccination; onset pattern acute/subacute. **Course:** episodic active disease during infection with recovery in survivors; survivors are typically asymptomatic after early childhood. **Critical period:** early infancy around BCG exposure is the window of vulnerability and of opportunity for early recognition/intervention. **Remission:** treatment-induced with antimycobacterial therapy ($\pm$ IFN- γ).

9. Inheritance and Population

Inheritance: X-linked recessive; affects males, mothers are obligate carriers. **Penetrance:** appears high in exposed hemizygous males but based on only 5 patients. **Expressivity:** variable (recovery to death). **Epidemiology:** ultra-rare — 5 reported patients worldwide; prevalence/incidence not estimable. **Populations:** four ancestries reported (Chinese, Finnish, Iranian, Saudi Arabian) — no single founder population. **Sex ratio:** essentially all affected are male (X-linked recessive). **Consanguinity:** relevant for autosomal recessive MSMD genes but less so here given X-linkage; the geographic spread suggests independent (private) mutations rather than a founder effect. **Carrier frequency:** unknown/very low.

10. Diagnostics

Routine immune workup is typically **normal** (normal circulating leukocyte subsets), so diagnosis rests on **molecular genetics**. Recommended approach: **whole-exome or whole-genome sequencing / targeted MSMD gene panels** including *MCTS1*, prompted by disseminated BCG/NTM disease in an otherwise healthy infant. Functional confirmation: reduced JAK2 protein, impaired IFN- γ after BCG stimulation, impaired IL-23 (but normal IL-12) responses, and **preserved responses to exogenous IFN- γ /IFN- α** — the last both diagnostic and therapeutically informative. Microbiology: blood/tissue mycobacterial culture and, increasingly, metagenomic next-generation sequencing (mNGS) to identify unculturable NTM. **Differential diagnosis:** other MSMD genotypes — IL12RB1, IL12B, IFNGR1/2, STAT1, ISG15, IRF8, IL23R, RORC, NEMO, CYBB — distinguished by their specific gene defects and, in some, broader infectious susceptibility.

11. Outcome / Prognosis

Generally favorable but potentially fatal: 1 of 5 patients died of disseminated mycobacterial disease; 4 recovered with antimycobacterial therapy, had no other infections, and showed normal growth and development. **Prognostic factors:** early diagnosis and prompt antimycobacterial ($\pm$ IFN- γ) therapy; intact response to exogenous IFN- γ is a favorable feature. Long-term survivors are typically asymptomatic after early childhood. Given the tiny cohort, survival statistics are indicative rather than definitive.

12. Treatment

Antimycobacterial therapy is the cornerstone (multidrug regimens directed at BCG/NTM). **Recombinant IFN- γ** is a rational adjunct because patient cells respond normally to it, bypassing the upstream block — a strategy with established benefit in other MSMD genotypes

(e.g., partial IFNGR1 and IL12RB1 defects). **Hematopoietic stem cell transplantation** is a consideration in severe/refractory MSMD generally, though not specifically reported for IMD118. No pharmacogenomic, gene, RNA, or targeted small-molecule therapies exist yet for MCTS1 deficiency. Suggested **NCIT** terms: recombinant interferon gamma; antimycobacterial agent; hematopoietic stem cell transplantation.

13. Prevention

Primary prevention: in families with a known *MCTS1* variant, **avoid live BCG vaccination** in at-risk male infants until immune status is clarified — BCG is the principal precipitant. **Secondary prevention:** early recognition of disseminated mycobacterial disease and prompt therapy. **Genetic counseling:** X-linked recessive inheritance — carrier mothers have a 50% chance of transmitting the variant; sons of carriers have a 50% risk of being affected, daughters a 50% chance of being carriers. **Carrier/prenatal testing** and cascade testing are available once the familial variant is known. **Tertiary prevention:** antimycobacterial prophylaxis and monitoring in affected individuals.

14. Other Species / Natural Disease

Species affected: **human (*Homo sapiens*, NCBI Taxon 9606)** — the disease is described only in humans. Mechanistic conservation is strong: the MCTS1/DENR-eIF2D recycling system has direct yeast counterparts (Tma20/Tma22/Tma64), demonstrating deep evolutionary conservation of the underlying translation machinery ([PMID: 30146315](#), [PMID: 38903097](#)). Orthologous *Mcts1* exists in mouse and other vertebrates. No naturally occurring animal disease (OMIA) is reported. No zoonotic component.

15. Model Organisms

No dedicated *Mcts1*-deficient mammalian disease model of IMD118 has been reported. **In vitro/cellular models** were central to the discovery: MCTS1-knockout HeLa cells (5'UTR reporter screens), CRISPR MCTS1-knockout THP-1 monocytes (endogenous JAK2 reduction across 5 clones), and patient-derived leukocytes. **Yeast** (*Saccharomyces cerevisiae*, Tma20/Tma22/Tma64) is a powerful model for the ribosome-recycling/reinitiation mechanism itself. These systems recapitulate the molecular defect (JAK2 translational block, ribosome stalling) but not the whole-organism immunophenotype; a conditional or humanized mouse expressing the *JAK2* stuORF-containing 5'UTR would be needed to model the immunological disease in vivo.

Evidence Base

PMID	Title (abbrev.)	Role in this report
37875108	<i>Human MCTS1-dependent translation of JAK2 is essential for IFN-γ immunity to mycobacteria</i>	Primary/defining source. Identifies IMD118, the 5 patients, variant spectrum, JAK2 stuORF mechanism, IL-23 $\rightarrow$ IFN- γ /MAIT- $\gamma\delta$ axis, outcomes (1 death), preserved exogenous IFN- γ response.
30146315	<i>Tma64/eIF2D, Tma20/MCT-1, Tma22/DENR recycle post-termination 40S subunits in vivo</i>	Establishes MCTS1/DENR/eIF2D as 40S recycling factors; evolutionary conservation.
38903097	<i>Impacts of yeast Tma20/MCTS1, Tma22/DENR and Tma64/eIF2D on translation reinitiation</i>	Shows MCTS1/DENR enhance reinitiation at uORFs — mechanism underlying JAK2 dependence.
16982740	<i>MCT-1 protein interacts with the cap complex...</i>	Defines MCTS1 PUA domain, cap-complex interaction, DENR recruitment.
35115540	<i>Cyclin B/CDK1 and Cyclin A/CDK2 phosphorylate DENR...</i>	Establishes MCTS1/DENR as oncogenes and cell-cycle-regulated uORF translation factors.
37634410 / 35274760	MCTS1 in laryngeal squamous cell carcinoma	Oncogenic role of MCTS1 (contrast with benign germline-loss phenotype).
36763636	<i>Human IL-23 is essential for IFN-γ-dependent immunity to mycobacteria</i>	Validates the IL-23 $\rightarrow$ IFN- γ axis that MCTS1/JAK2 loss disrupts.
35829840	<i>Homozygous stop mutation in IL23R causes MSMD</i>	Independent genetic validation that disrupting IL-23 signaling causes MSMD.
37147816 / 41707313	MAIT cells in mycobacterial immunity	Effector-cell biology: MAIT cells produce IFN- γ via MR1, reduced in active TB.
40656276 / 41786143 / 41048447	MSMD overviews & management	Context for MSMD classification, diagnostics, and IFN- γ /antimycobacterial/HSCT management.

Evidence source types: IMD118-defining data are **human clinical + in vitro/cellular** (Bohlen 2023); mechanistic recycling/reinitiation data are **in vitro + model organism (yeast)**; pathway validation is **human genetic** (IL23R, IL-23); oncogenic role is **in vitro / tumor genomics**.

Limitations and Knowledge Gaps

1. **Single defining cohort (n = 5).** Nearly all disease-specific knowledge derives from one report (PMID: 37875108). Prevalence, penetrance, expressivity, sex-specific carrier phenotypes, and long-term outcomes are therefore uncertain.
2. **No formal epidemiology.** Prevalence and incidence cannot be estimated; the disease is ultra-rare and likely underdiagnosed.
3. **No animal model of the disease.** Mechanistic work used cell lines and yeast; whole-organism recapitulation (e.g., stuORF-humanized mouse) is absent, so in vivo immunodynamics remain inferred.
4. **Therapeutic evidence is indirect.** IFN- γ benefit is inferred from preserved in vitro responses and from other MSMD genotypes, not from a trial in MCTS1-deficient patients.
5. **Spectrum of mycobacterial species and non-mycobacterial risk** is incompletely defined given the small cohort; whether MCTS1-deficient patients have subtle susceptibility to other intracellular pathogens (as some MSMD genotypes do) is unknown.
6. **Female carriers / X-inactivation effects** have not been characterized.
7. **Cancer risk over the lifespan** is unassessed — germline MCTS1 loss is oncologically "silent" in childhood, but the long-term consequences of losing an oncogene's function are unstudied.

Proposed Follow-up Experiments / Actions

1. **International case-finding and registry.** Add *MCTS1* to MSMD/IEI sequencing panels worldwide and pool cases (e.g., via GeneMatcher) to define penetrance, expressivity, microbial spectrum, and outcomes.
2. **Therapeutic evaluation of recombinant IFN- γ** as adjunct to antimycobacterial therapy in confirmed patients, leveraging their intact IFN- γ responses; document response rates and adverse events.

3. **Generate an in vivo model** — a knock-in mouse humanizing the *JAK2* 5'UTR stuORFs on an *Mcts1*-deficient background, or a conditional myeloid/T-cell *Mcts1* knockout — to test whether the JAK2→IL-23→IFN- γ chain and mycobacterial susceptibility are recapitulated.
4. **Ribosome profiling** in patient-derived macrophages and MAIT/ $\gamma\delta$ cells to catalog the full set of MCTS1-dependent (stuORF/uORF) transcripts beyond JAK2 and confirm JAK2's privileged dependence.
5. **Vaccination guidance:** formalize the recommendation to withhold live BCG in at-risk male infants of carrier mothers pending genetic clarification.
6. **Long-term surveillance** of survivors for late infections and — given MCTS1's oncogene status — for any oncologic outcomes, to establish lifetime prognosis.
7. **Structural/biophysical work** on how patient variants (esp. p.Ala133_Lys181del) disrupt MCTS1–DENR–ribosome interactions, to inform potential small-molecule or read-through therapeutic strategies.

Report compiled from 7 confirmed findings and 35 reviewed papers over 5 investigation iterations. All disease-specific clinical and mechanistic claims trace primarily to Bohlen et al. 2023 (PMID: 37875108); mechanistic and pathway claims are corroborated by the cited translation-machinery and MSMD literature.
